Semester project

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Introduction

Governments around the world are introducing policies to reduce greenhouse gas (GHG) emissions and support the transition to a lower-carbon economy. Carbon taxes are among the most important of these tools, and Europe is where carbon pricing is most developed. Finland introduced the world's first carbon tax in 1990, starting at a low rate that rose over time, and the policy has since inspired other nations to consider similar approaches (EARTH.ORG). Today, 24 European countries have a carbon tax. By putting a price on carbon, the tax makes emitting more costly and gives households and firms a reason to cut back.

Carbon tax rates vary enormously. In 2024, they ranged from about US$0.08 per tonne of CO2 in Poland to US$124 in Sweden, and some European countries have no carbon tax at all. This raises a natural question: do carbon taxes actually reduce emissions? At first glance, countries with higher taxes should emit less. However, wealthy countries and countries with a large share of low-carbon energy may have low emissions with or without a tax. We therefore control for income and the energy mix before drawing any conclusions.

Our research question is:

Does a high carbon tax rate lead to lower GHG emissions per capita after factoring in fossil fuel share in the energy mix and GDP?

To answer it, we use a panel of 22 European countries over 2005-2024, combining carbon tax rates from the World Bank with emissions, population, GDP and energy data from Eurostat. The analysis shows associations rather than causal effects, since governments with stronger climate ambition tend to set higher tax rates.

Data sources

Cleaning

The raw files we found contains many variables and columns we don't need. Therefore, for each file we only kept the country, year and the value. The final dataset was built in the following steps:

1. **Country and year:** We kept the 22 countries that have data in all five sources. Covering the period 2005-2024, which results in 420 country-year observations.
2. **Carbon tax rate:** The data source from The World Bank has one row per tax and one column per year. We only kept the instruments of type "carbon tax" were we matched each to its country. The dataset was restructured into panel format, with one observation per country and year. An empty cell means that no carbon tax existed in that country in that year, so we set it to 0.
3. **Countries without a national carbon tax:** Austria, Belgium, Germany, Greece, Italy and Lithuania don't have rows in the carbon tax data. The reason is that they price carbon only through the EU Emissions Trading Systems. We code them as 0 in all years.
4. **EU ETS is not included:** The ETS price is the same for all EU countries in a given year. Therefore it doesn't add a variation between the EU countries.
5. **Emissions per capita**: Eurostat reports total GHG emissions in million tonnes. We therefore multiply by 1 000 000 and divide by population to get tonnes of CO2 per person.
6. **Fossil fuel share:** For each country and year, we add up solid fossil fuels, natural gas, oil and petroleum products. These are divided by total gross available energy. We then take the denominator from Eurostats "Total" row instead of summing the other categories ourselves, si that its matches Eurostats official total.
7. **Units:** The carbon tax is measured in US dollars and the GDP per capita in euros, and we don't convert one into the other. This doesn't affect the regression, because the two variables enter seperatley. What it does is that the tax coefficient is interpreted per US dollars per tonne of CO2, not per euro.

Descreptive statistics

Explain what the data shows us from the final dataset. ow many countries ever had a carbon tax and how many never did, the range of each variable, and anything unusual (for example Luxembourg's high emissions in the early years). Mention the correlations Half a page

Exploring the data

RQ:

Does a high carbon tax rate lead to lower GHG emissions per capita after factoring in fossil fuel share in the energy mix and GDP?

packages installed:

install.packages("tidyverse") install.packages("tinytex") install.packages("hrbrthemes") install.packages("ggrepel")

```
library(tidyverse)
library(readxl)
library(ggplot2)
library(hrbrthemes)
library(ggrepel)

Carbon_tax <- read_excel("Carbon_tax.xlsx", sheet = "Compliance_Price", skip = 1)
GHG <- read_csv("GHG.csv")
Population <- read_csv("Population.csv")
GDP <- read_csv("GDP.csv")
Energy_mix <- read_csv("Energy_mix.csv")
```

(Tax rate is USD per tonne of CO2-equivalent)

```
#Choosing and renaming relevant columns
My_countries <- c("Austria", "Belgium", "Denmark", "Estonia", "Finland",
                  "France", "Germany", "Greece", "Hungary", "Iceland",
                  "Ireland", "Italy", "Latvia", "Lithuania", "Luxembourg",
                  "Netherlands", "Norway", "Poland", "Portugal",
                  "Slovenia", "Spain", "Sweden")

Year_range <- 2005:2024

Carbon_tax_clean <- Carbon_tax %>%
  filter(
    `Instrument Type` == "Carbon tax"
    ) %>% #filtering out ETS only keeping carbon tax
  mutate(
    Country = str_extract(`Name of the initiative`,
  paste(My_countries, collapse = "|"))
  ) %>%
  filter(!is.na(Country)) %>%
  select(
    Country, all_of(as.character(Year_range))
    ) %>%
  pivot_longer(
    cols = -Country, names_to = "Year", values_to = "Tax_rate_usd"
```

```
    ) %>%
  mutate(
    Year = as.numeric(Year),
    Tax_rate_usd = ifelse(is.na(Tax_rate_usd), 0, Tax_rate_usd)
    )

no_tax_countries <- setdiff(My_countries, unique(Carbon_tax_clean$Country))

no_tax_filled <- expand_grid(Country = no_tax_countries, Year = Year_range
    ) %>%
  mutate(Tax_rate_usd = 0)

Carbon_tax_clean <- bind_rows(Carbon_tax_clean, no_tax_filled) %>%
  arrange(Country, Year)

#Choosing and renaming relevant columns
GHG_clean <- GHG %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
    ) %>%
  rename(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    Total_emissions = OBS_VALUE
  )

#Choosing and renaming relevant columns
Population_clean <- Population %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
  ) %>%
  rename(Country = `Geopolitical entity (reporting)`,
         Year = TIME_PERIOD,
         Population = OBS_VALUE
         )
#Choosing and renaming relevant columns
GDP_clean <- GDP %>%
  select(
    `Geopolitical entity (reporting)`, TIME_PERIOD, OBS_VALUE
  ) %>%
  rename(
    Country = `Geopolitical entity (reporting)`,
```

```
    Year = TIME_PERIOD,
    GDP_per_capita = OBS_VALUE
  )

#Choosing and renaming relevant columns
Energy_mix_clean <- Energy_mix %>%
  select(
    Country = `Geopolitical entity (reporting)`,
    Year = TIME_PERIOD,
    Source = `Standard international energy product classification (SIEC)`,
    OBS_VALUE
  ) %>%
  filter(Source %in% c("Solid fossil fuels",
  "Natural gas",
  "Oil and petroleum products (excluding biofuel portion)", "Total")
         ) %>%
  pivot_wider(
    names_from = Source, values_from = OBS_VALUE, values_fill = 0
    ) %>%
  mutate(
    Fossil_fuels_share = (`Solid fossil fuels` + `Natural gas` +
    `Oil and petroleum products (excluding biofuel portion)`)
    / Total * 100
    ) %>%
  select(
    Country, Year, Fossil_fuels_share
    )
```

```
Merged1 <- merge(
  Carbon_tax_clean, GHG_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Merged2 <- merge(
  Population_clean, GDP_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )

Merged3 <- merge(
  Merged1, Merged2,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )
```

```r
Trial_dataset <- merge(
  Merged3, Energy_mix_clean,
  by.x = c("Country","Year"), by.y = c("Country", "Year")
  )
```

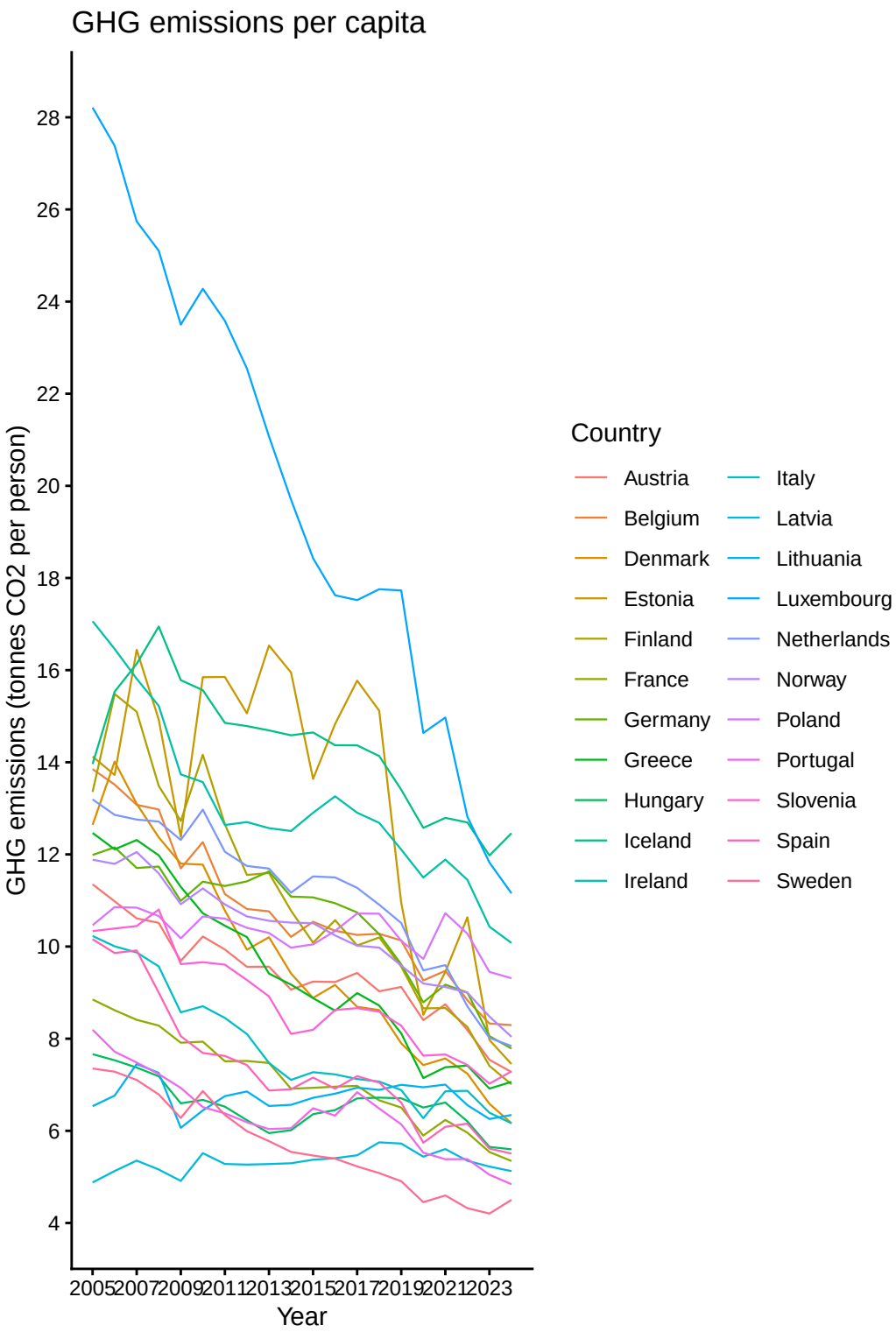
Unit of measure is in tonnes

```r
Trial_dataset$Emissions_per_capita <-
  Trial_dataset$Total_emissions * 1000000 / Trial_dataset$Population

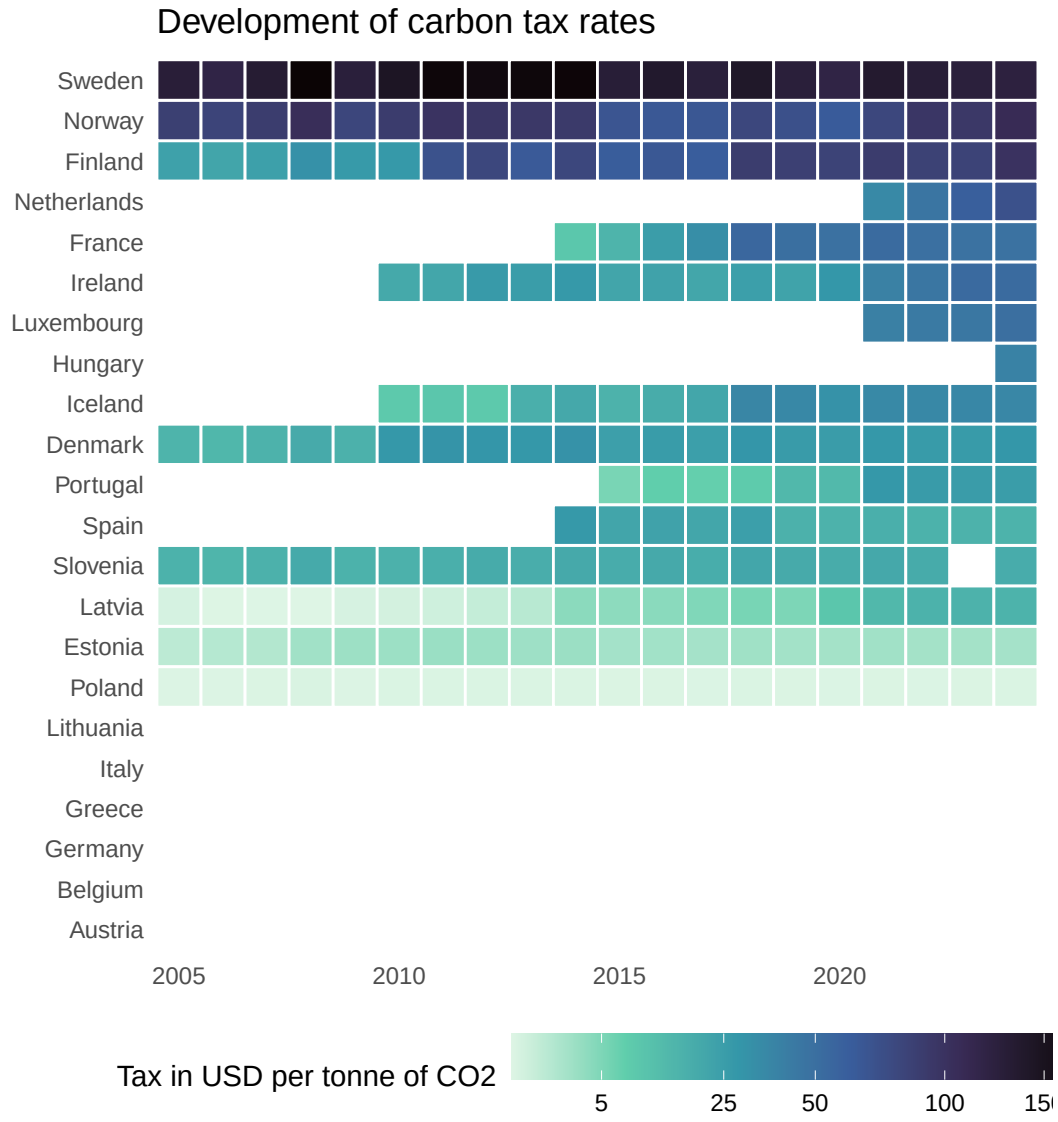
Final_dataset <- Trial_dataset %>%
  select(
    Country, Year, Tax_rate_usd, GDP_per_capita, Fossil_fuels_share,
    Emissions_per_capita
  )
```

Plotting each countrie GHG emissions per capita throughout the years. How did emissions evolve throughout the years?

```r
ggplot(
  Final_dataset, aes(x = Year, y = Emissions_per_capita, color = Country)) +
  geom_line(linewidth = 0.4) +
  scale_x_continuous(breaks = seq(2005, 2024, by = 2)) +
  scale_y_continuous(breaks = seq(0, 30, by = 2)) +
  labs(
    title = "GHG emissions per capita", y = "GHG emissions (tonnes CO2 per person)"
  ) +
  theme_classic(base_size = 11)
```


GHG emissions per capita
GHG emissions (tonnes CO2 per person)
28
26
24
22
20
18
16
14
12
10
8
6
4
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
Year
Country
Austria
Belgium
Denmark
Estonia
Finland
France
Germany
Greece
Hungary
Iceland
Ireland
Italy
Latvia
Lithuania
Luxembourg
Netherlands
Norway
Poland
Portugal
Slovenia
Spain
Sweden

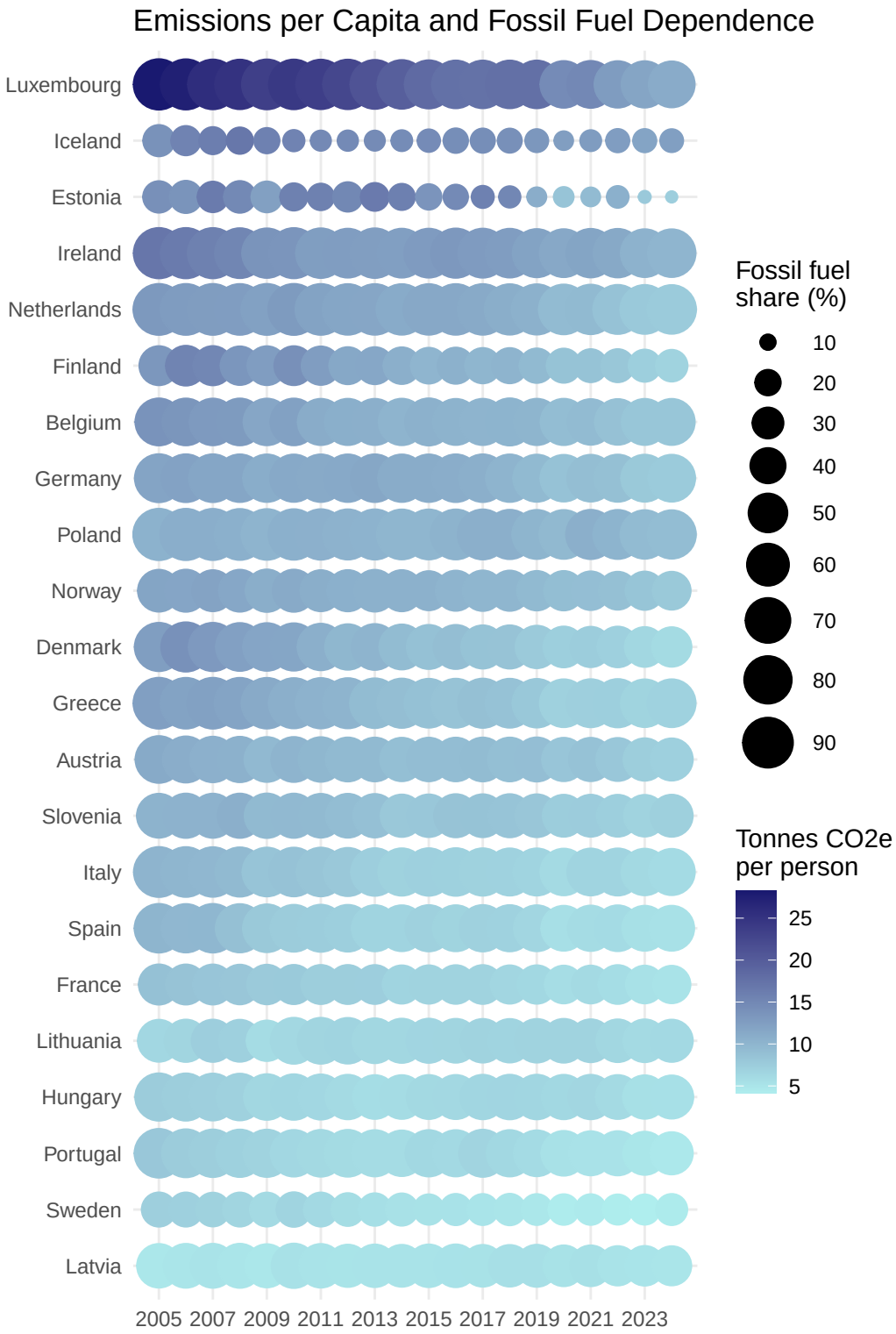
```r
Final_dataset %>%
   mutate(
    Carbon_Tax = ifelse(Tax_rate_usd <= 0, NA, Tax_rate_usd),
    Country = fct_reorder(Country, Tax_rate_usd, .fun = max)
  ) %>%
  ggplot(aes(Year, Country, fill = Carbon_Tax)) +
  geom_tile(colour = "white", linewidth = 0.5) +
   theme_minimal(base_size = 11) +
  scale_x_continuous(breaks = seq(2005, 2024, by = 5), expand = c(0, 0)) +
  labs(x = NULL, y = NULL) +
    theme(panel.grid = element_blank(), legend.position = "bottom", legend.key.width = unit(
  ) +
  scale_fill_viridis_c(
    option = "mako", direction = -1, trans = "sqrt",
    breaks = c(5, 25, 50, 100, 150),
    na.value = "white",
    name = "Tax in USD per tonne of CO2"
  ) +
  labs(title = "Development of carbon tax rates")
```



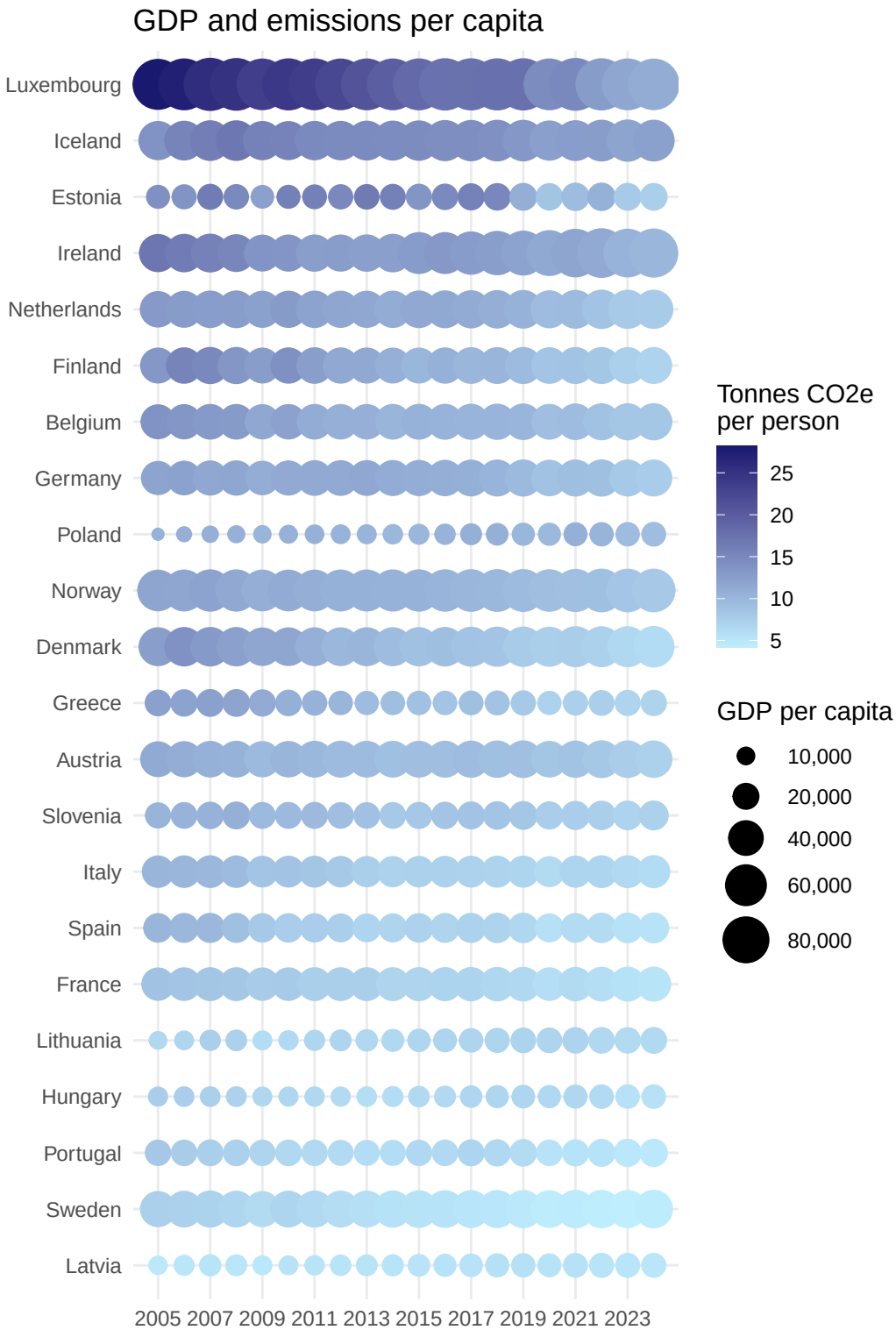
#PLotting fossil fuels share in the countries energy mix, and in a seperte plot emissions agains GDP.

```
Final_dataset %>% #Putting the countries in order from highest to lowest e, and getting the a
  mutate(Country = reorder(Country, Emissions_per_capita, FUN = mean)) %>%
  ggplot(aes(Year, Country, size = Fossil_fuels_share, colour = Emissions_per_capita)) +
  geom_point(alpha = 1) +
  scale_colour_gradient(low = "paleturquoise", high = "midnightblue", breaks = seq(0, 30, by
  scale_size(range = c(2, 10), breaks = seq(0, 100, by = 10),
             name = "Fossil fuel\nshare (%)") +
```

```
  scale_x_continuous(breaks = seq(2005, 2024, by = 2)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(), legend.position = "right") +
  labs(title = "Emissions per Capita and Fossil Fuel Dependence")
```


Emissions per Capita and Fossil Fuel Dependence
Luxembourg
Iceland
Estonia
Ireland
Netherlands
Finland
Belgium
Germany
Poland
Norway
Denmark
Greece
Austria
Slovenia
Italy
Spain
France
Lithuania
Hungary
Portugal
Sweden
Latvia
2005 2007 2009 2011 2013 2015 2017 2019 2021 2023
Fossil fuel share (%)
10
20
30
40
50
60
70
80
90
Tonnes CO2e per person
25
20
15
10
5

```r
#Order of countries by average emissions (lowest at bottom, highest at top)
Final_dataset %>%
  mutate(Country = reorder(Country, Emissions_per_capita, FUN = mean)) %>%
  ggplot(aes(Year, Country, size = GDP_per_capita, colour = Emissions_per_capita)) +
  geom_point(alpha = 5) +
  scale_colour_gradient(low = "lightblue1", high = "midnightblue", breaks = seq(0, 30, by = 5
  scale_size(range = c(2, 10), breaks = c(10000, 20000, 40000, 60000, 80000, 110000), labels
             name = "GDP per capita") +
  scale_x_continuous(breaks = seq(2005, 2024, by = 2)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(), legend.position = "right") +
  labs(title = "GDP and emissions per capita")
```


GDP and emissions per capita
Luxembourg
Iceland
Estonia
Ireland
Netherlands
Finland
Belgium
Germany
Poland
Norway
Denmark
Greece
Austria
Slovenia
Italy
Spain
France
Lithuania
Hungary
Portugal
Sweden
Latvia
2005 2007 2009 2011 2013 2015 2017 2019 2021 2023
Tonnes CO2e per person
25
20
15
10
5
GDP per capita
10,000
20,000
40,000
60,000
80,000